

Predicting Paralog Compensation Before Editing a Soybean Gene

A multi-evidence computational pipeline, validated on the soybean
leghemoglobin / hemoglobin gene family

Expression coexpression · Molecular evolution (aBSREL) · Protein language model (ESM-2)

Single-gene edits in soybean often produce no phenotype

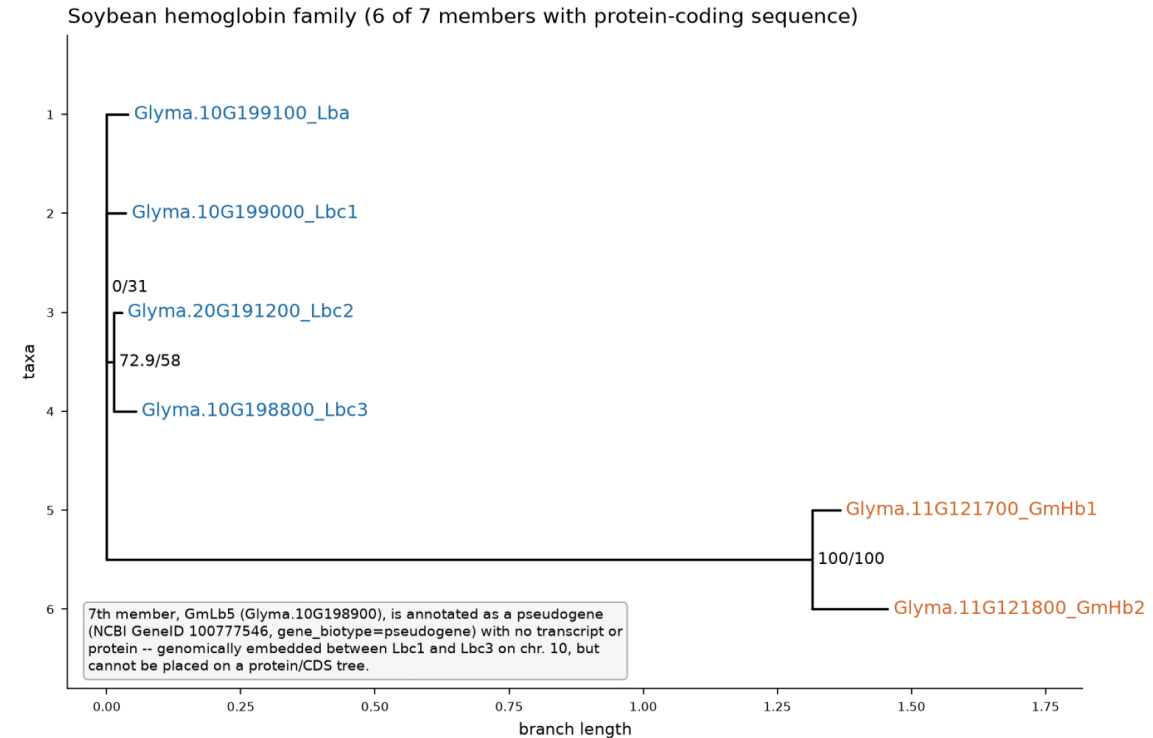
- CRISPR knockouts routinely fail to show an expected phenotype in polyploid/duplicated crop genomes.
- Soybean underwent two rounds of whole-genome duplication (~59 Mya and ~13 Mya, Schmutz et al. 2010), and retains a very large fraction of its genes in duplicate.
- When a gene has one or more paralogs, those paralogs can mask the loss of the edited copy through functional redundancy -- a phenomenon variously called genetic buffering, dosage compensation, or transcriptional adaptation.
- Result: wasted breeding/editing cycles, false-negative phenotyping, and wasted greenhouse time when a single-gene edit was never going to show an effect on its own.

Can we predict compensation risk before making the edit?

- If a target gene's paralog(s) are likely to compensate for its loss, a single-gene edit should be planned as part of a multiplex/combinatorial knockout from the start.
- If the paralog(s) are functionally diverged, a single-gene edit is informative on its own -- no extra editing burden needed.
- Today this call is usually made by intuition, sequence identity alone, or after the fact once a single-gene edit has already failed to show a phenotype.
- This work builds and validates a computational pipeline that scores compensation risk from three independent lines of evidence, before any wet-lab work.

The soybean hemoglobin family: 7 genes, 2 clades

- Symbiotic leghemoglobins (nodule O₂ buffering):
 - Lba, Lbc1, Lbc2, Lbc3 + GmLb5 (pseudogene, no protein) -- Glyma IDs on the tree
- Non-symbiotic hemoglobins (distinct roles):
 - GmHb1 (Glyma.11G121700), GmHb2 (Glyma.11G121800)
- One family, two subclades -- a within-family test
 - bed for redundant vs. diverged pairs.



Three independent lines of evidence, per paralog pair

1. Expression coexpression

- Map paralogs onto a single-cell/tissue expression atlas (GSE270392, soybean).
- Classify each pair by rank-standardized coexpression and fold-change dispersion (Benoit et al. 2025 framework: dosage-balanced / dominant / specialized / diverged).
- Captures: do the two genes get used in the same cells at the same time?

2. Molecular evolution

- Build a maximum-likelihood tree (MAFFT + IQ-TREE) and run HyPhy aBSREL branch-site tests.
- A paralog under relaxed purifying selection (elevated omega on its branch) is a signature of neofunctionalization -- evolving away from redundancy.
- Captures: has this copy been evolving under different constraints since duplication?

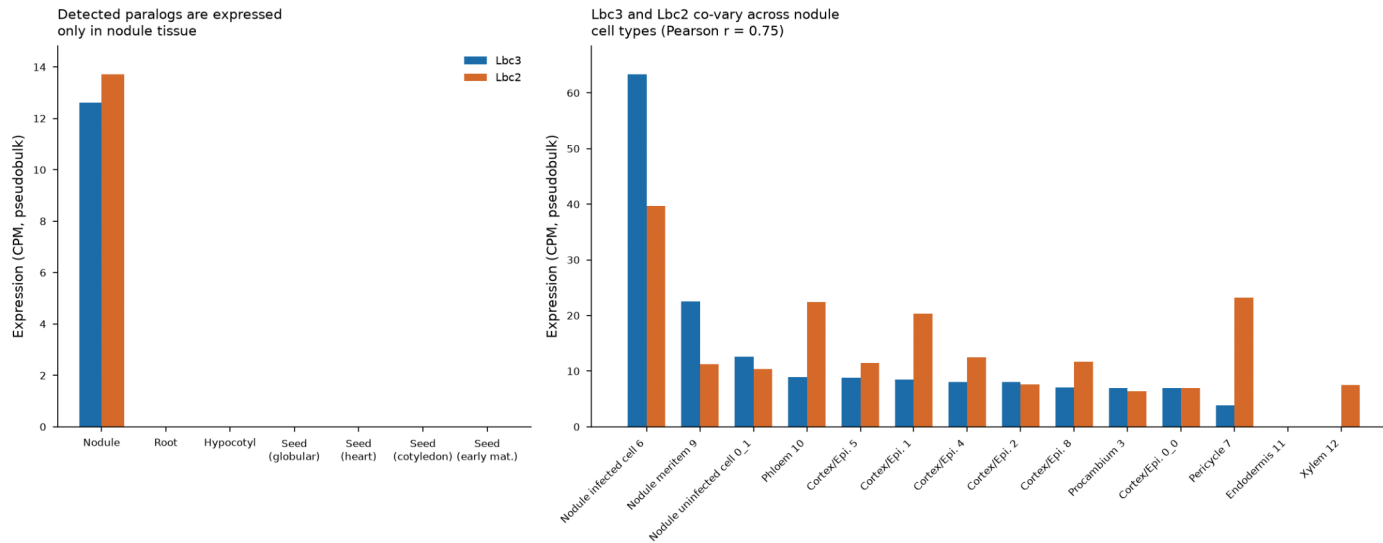
3. Protein language model

- Score every shared alignment column with ESM-2 masked-marginal predictions.
- Compensation likelihood = how plausible the model finds each paralog's residues in the other's sequence context, averaged over both directions.
- Captures: could one paralog's protein still fold/function wearing the other's residues?

Pipeline: fetch → tree → expression → selection → PLM → report

- Fetch OrthoDB orthologous-group sequences, or a literature-curated gene set (as for GmAS1/GmAS2, GmHb1/GmHb2 here).
- Build a MAFFT alignment + IQ-TREE maximum-likelihood tree (any species; dispatched to Modal).
- For soybean paralogs: resolve Glyma IDs, map onto the GSE270392 expression atlas, compute Benoit et al. classification metrics.
- Run HyPhy aBSREL branch-site selection (soybean subtree and the full multi-species tree) on Modal.
- Score ESM-2 paralog compensation likelihood for every same-species pair, on Modal.
- Packaged as a published Claude skill (orthodb-paralog-pipeline) with reusable code.

Lbc3 vs. Lbc2: all lines of evidence agree on redundancy



- Expression: both nodule-specific, coexpression = 0.945 → Benoit et al. Group I (dosage-balanced).
- Selection: no significant aBSREL signal on either branch.
- PLM: ESM-2 compensation score = +0.0015 -- near-perfect mutual substitutability.
- Prediction: HIGH compensation risk.

GmHb1 vs. GmHb2: a divergent pair, same family

- GmHb1 and GmHb2 (non-symbiotic hemoglobins) have documented, non-overlapping expression domains
 - leaves vs. roots, respectively (Du et al. 2020) -- the opposite of Lbc3/Lbc2's shared nodule expression.
- Testing whether the pipeline correctly calls this pair LESS mutually substitutable than Lbc3/Lbc2: lower ESM-2 compensation score, and/or a detectable aBSREL selection signal.
- [PENDING -- results being computed on Modal now; this slide will be updated with the compensation score, aBSREL result, and comparison figure.]

Does the pipeline's signal hold up beyond one family?

- Cross-metric concordance: do expression, selection, and PLM compensation scores agree with each other across a wider panel of soybean paralog pairs (leghemoglobin family + the literature-documented GmAS1/GmAS2 leaf-shape gene family, 36 pairs total)?
- Null-distribution test: are true within-family paralog compensation scores distinguishable from a null built on random cross-family (unrelated) sequence pairs?
- WGD-age stratification: paralog pairs from soybean's more recent whole-genome duplication (~13 Mya, low Ks) versus the older event (~59 Mya, higher Ks) -- does compensation likelihood track duplication age, as expected if it reflects real divergence?
- [PENDING — three benchmark analyses running now; this slide will be updated with correlation statistics, empirical p-values, and figures.]

A decision rule for planning an edit

Before committing to a single-gene edit, run this pipeline on the target's paralog family:

► HIGH compensation risk

High coexpression + near-zero PLM compensation score + no selection signal
→ plan a multiplex/combinatorial knockout of the paralog group from the start.

► LOW compensation risk

Divergent expression domains + lower PLM compensation score + detectable selection signal
→ a single-gene edit is likely to be informative on its own.

This reframes paralog compensation from a post-hoc explanation for a failed edit into a pre-registered, quantitative prediction.

What this does not yet show

- The ESM-2 compensation score is a sequence/structural-plausibility proxy, not a functional assay -- it has not been validated against real CRISPR knockout phenotypes in soybean.
- Benchmark panels here are small (2 gene families, 36 pairs) -- broader validation needs a larger, more diverse panel of soybean paralog families with known editing outcomes.
- Ks-based WGD-age binning breaks down at high divergence (NG86 saturates); cross-subfamily pairs with dS at or above ~ 1.5 are excluded from that specific comparison.
- Next step: validate against published soybean CRISPR studies with explicit single- vs. multi-gene knockout phenotype comparisons (e.g. GmAS1 quadruple mutants) as ground truth.